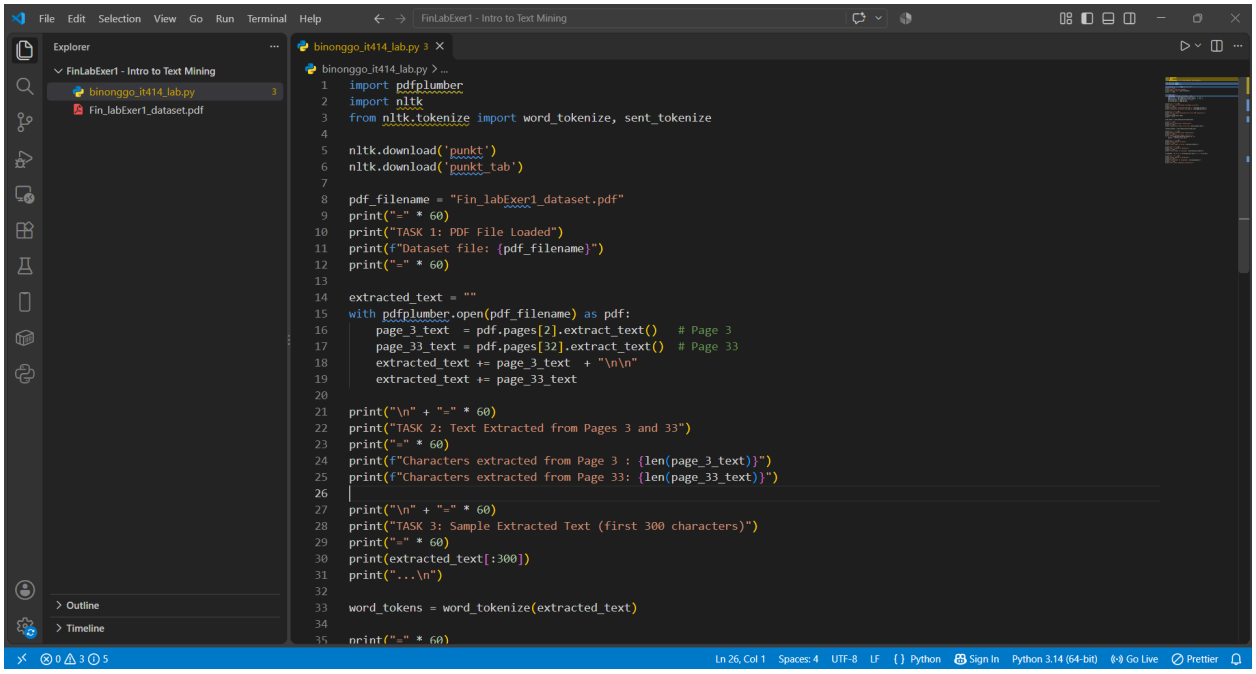
Republic of the Philippines
UNIVERSITY OF SCIENCE AND TECHNOLOGY OF SOUTHERN PHILIPPINES
Alubijid | Balubal | Cagayan de Oro | Claveria | Jasaan | Oroquieta | Panaon | Villanueva

COLLEGE OF INFORMATION TECHNOLOGY AND COMPUTING
Department of Information Technology

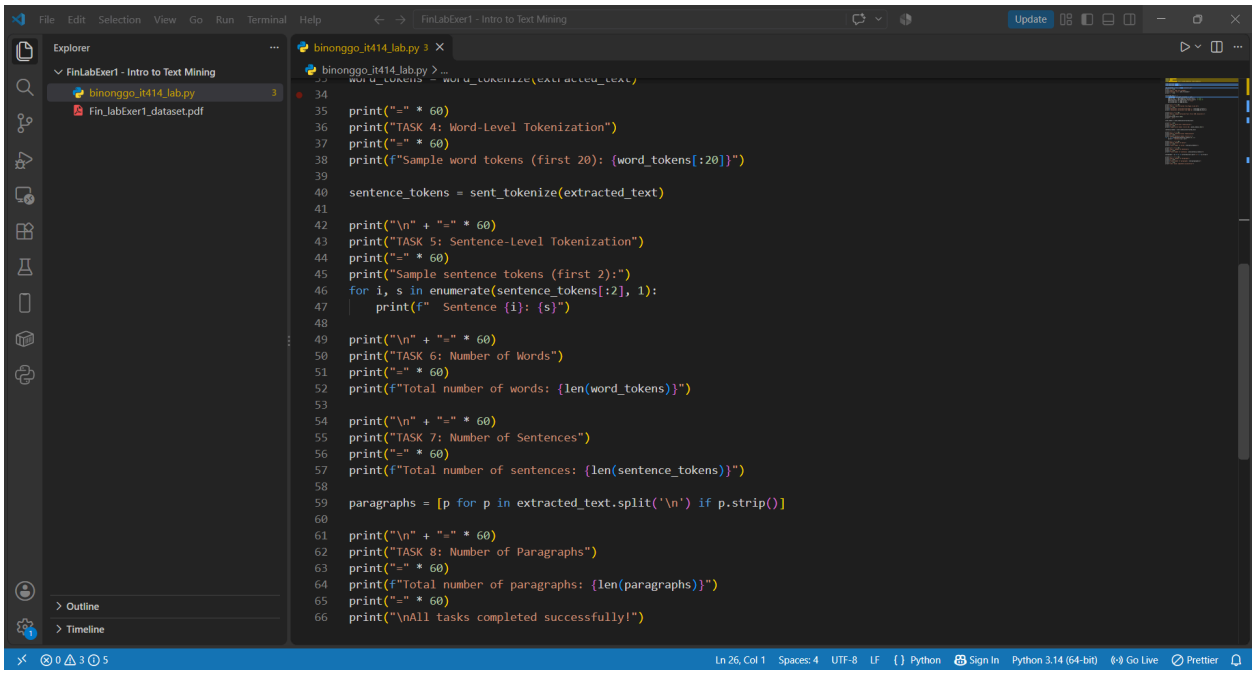
Name: John Michael Angelo C. Binonggo
Date: 10-25-2026
Program and Section: BSIT-4R8
Score: _____

FinLabExer1 - Intro to Text Mining

Code Screenshot via VSCode:



Screenshot 1



Screenshot 2

Code:

```
import pdfplumber
import nltk
from nltk.tokenize import word_tokenize, sent_tokenize


nltk.download('punkt')
nltk.download('punkt_tab')


pdf_filename = "Fin_labExer1_dataset.pdf"
print("=" * 60)
print("TASK 1: PDF File Loaded")
print(f"Dataset file: {pdf_filename}")
print("=" * 60)
```

```
extracted_text = ""
with pdfplumber.open(pdf_filename) as pdf:
    page_3_text  = pdf.pages[2].extract_text()   # Page 3
    page_33_text = pdf.pages[32].extract_text()  # Page 33
    extracted_text += page_3_text  + "\n\n"
    extracted_text += page_33_text

print("\n" + "=" * 60)
print("TASK 2: Text Extracted from Pages 3 and 33")
print("=" * 60)
print(f"Characters extracted from Page 3 : {len(page_3_text)}")
print(f"Characters extracted from Page 33: {len(page_33_text)}")

print("\n" + "=" * 60)
print("TASK 3: Sample Extracted Text (first 300 characters)")
print("=" * 60)
print(extracted_text[:300])
print("...\n")

word_tokens = word_tokenize(extracted_text)

print("=" * 60)
print("TASK 4: Word-Level Tokenization")
print("=" * 60)
print(f"Sample word tokens (first 20): {word_tokens[:20]}")

sentence_tokens = sent_tokenize(extracted_text)
```

```
print("\n" + "=" * 60)
print("TASK 5: Sentence-Level Tokenization")
print("=" * 60)
print("Sample sentence tokens (first 2):")
for i, s in enumerate(sentence_tokens[:2], 1):
    print(f"  Sentence {i}: {s}")

print("\n" + "=" * 60)
print("TASK 6: Number of Words")
print("=" * 60)
print(f"Total number of words: {len(word_tokens)}")

print("\n" + "=" * 60)
print("TASK 7: Number of Sentences")
print("=" * 60)
print(f"Total number of sentences: {len(sentence_tokens)}")

paragraphs = [p for p in extracted_text.split('\n') if p.strip()]

print("\n" + "=" * 60)
print("TASK 8: Number of Paragraphs")
print("=" * 60)
print(f"Total number of paragraphs: {len(paragraphs)}")
print("=" * 60)
print("\nAll tasks completed successfully!")
```

Personal Reflection:

Through this activity, I learned how text mining works and how Python can help process information from PDF documents. At first, I thought extracting text from a PDF was simply about getting the words from a document, but I realized that there are several steps involved before the data can be properly analyzed.

Using pdfplumber, I was able to extract the text from my assigned pages (Pages 3 and 33). I then used NLTK to perform word-level and sentence-level tokenization, which helped me understand how a large amount of text can be broken down into smaller and more organized parts. Based on the results, the extracted pages contained 1,008 words, 41 sentences, and 90 paragraphs.

What I found interesting was how Python could perform these tasks in just a few lines of code. It made me realize that text mining is not only about collecting information but also about preparing and organizing unstructured data so that it becomes easier to understand and analyze.

This activity gave me a better understanding of the basic process of Natural Language Processing (NLP). It also showed me how important text extraction and preprocessing are as starting points for more advanced applications, such as text analytics and sentiment analysis. I can see how these techniques can be useful when working with large amounts of textual data, where manually reading and analyzing everything would take much more time and effort.